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RESEARCH ARTICLE

Museum architecture as spatial storytelling of historical time: Manifesting a primary example of Jewish space in Yad Vashem Holocaust History Museum

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Abstract
Museums commonly adopt storytelling in their interpretive framework by use of audiovisual techniques to convey the meanings contained within artifacts. In addition to audiovisual mediation, this study demonstrates the idea that museum architecture itself can also be regarded as a medium of spatial storytelling, specifically of historical time, which is manifested spatially and cognitively for museum visitors.
The Yad Vashem Holocaust History Museum (YVHHM) in Jerusalem is considered a spatial storytelling tool that successfully establishes an architectural dimension and thus displays, reveals, and interprets historical time during the Holocaust. The research method of this study is drawn from a case study of YVHHM and consists of a literature review of scholarship in museum studies about artifacts and exhibition techniques of storytelling.
The study concludes that the architectural space and landscape of YVHHM create a primary example of Jewish space and its specific engagements with historical time by use of spatial layout and circulation, spatial form and symbolization, and spatial qualities of lighting and material. These components construct a tangible, sacred, and cultural artifact; such artifact inherits, preserves, and records Yad Vashem, Modern Jerusalem, and the Nation of Israel and is an ideal physical and spiritual "home" for Jewish people worldwide.
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1. Introduction

> If museums are to have a cultural role as distinct from that of the theme park, it lies in helping us orient ourselves and make discoveries in a world in which inherited common-sense conceptions of time and place are increasingly redundant (Lumly, 1988, p.18).

Human beings live in a material world; that is, they wear clothes and eat food. Man-made artifacts, from tiny pieces of jewelry to giant buildings, connect humans together as a society. Museum architecture collects significant artifacts within itself and thus occupies a dominant position in the contemporary era. Regardless of generating debate in the academic arena or converging foci in the field of practice, museum architecture is distinguished by researchers from other types of architecture, owing to its social significance for interpreting and mediating human history, culture, and civilization by conveying significant meanings contained in artifacts. Section 2 adopts storytelling in its daily display routine for interpretation and mediation and offers a historical overview of storytelling in museum architecture from early modern to postmodern societies, which consists of scholarship in museum studies about artifacts and exhibition techniques of storytelling. Section 3 selects the Yad Vashem Holocaust History Museum (YVHHM) as a particular case study of a spatial storytelling tool as a primary example of Jewish space. By revealing and manifesting the *historical time* of the Holocaust by use of spatial layout and circulation, spatial form and symbolization, and the spatial qualities of lighting and material, the idea of spatial storytelling contributes toward a unique embodied experience for the general public to support the process of "self-learning," as well as interpreting and mediating memory through tangible artifacts and architecture. Section 4 elaborates the conclusions of the study.

2. Storytelling in museum architecture: A historical overview

2.1. From collecting to self-learning and interpreting

Prior to providing an overview of storytelling in museums, a brief review of museum transformation should be made to contribute to the underlying reason that drives storytelling to be adopted in the daily display routine of museums; that is, the achievement of the social function of museums to mediate meanings with the general public.

In museums, the distinction between natural or cultural themes, represented by objects or artifacts in museum collections, must be abandoned (Dudley, 2009, p.xvi). The notion of nature as the isolated island of matter waiting for humans to peel the shell through the application of culture will no longer serve (p.xvi). The world is not a coin of raw material as opposed to constructed material goods but "rather a complex continuity of material relationships running from our bodies across the world, which are variously constructed into meanings of different kinds, of which 'nature' is one (p.xvi) and of which 'culture' is another. Culture is neither a universe in parallel with nature nor does it sojourn ambiguously in our minds. "Culture is created continually as we material beings engage with our material surroundings to produce the individual and social habits that add up to ongoing life" (p.xvi). Thus, the term "artifact" does not only refer to a thing made by human beings in a narrow sense but also refers to any displayed object in museums in a material sense.

In the pre-Enlightenment period, early museum collections began as private demonstrations by wealthy individuals or families and could be regarded as particular places for the rich to present their wealth to the general public and to preserve their reputations. Displayed in cabinets of curiosities, the content of collections varied from rare or curious objects d'art to natural objects and man-made artifacts. As products of the Enlightenment, the first public museums as "displays of artifacts for the edification and entertainment of the public" (Lumly, 1988, p.3) opened in Europe during the 18th century. According to observations by René Huyghe (1906–1997), a French writer on the history, psychology, and philosophy of art, the public museum and printed encyclopedia appeared at about the same time. For Kenneth Hudson (1916–1999), an industrial archaeologist, museologist, broadcaster, and author, the public museum and printed encyclopedia could be regarded as expressions of the 18th-century spirit of the Enlightenment, which produced "an enthusiasm for equality of opportunity of learning" (Hein, 1998, p.3). These movements were driven by the simple idea that a collection "which has hitherto been reserved for the pleasure and instruction of a few people should be made accessible to everybody" (Hudson, 1975, p.6).

According to George E. Hein, Professor at Lesley College, Cambridge, USA, the development of public museums in the 19th century can be divided into two stages. In the early stages of the 19th century, collections were focused on displays of "imperial conquests, exotic material, and treasures brought back to Europe by colonial administrations and private travellers or unearthed by increasingly popular excavations" (Hein, 1998, p.3) and were only open to those who were "fortunate enough to be allowed to enter and observe the splendo[*u*]r of a nation's wealth" (p.4). In the latter stages of the 19th century, museums were viewed as one of several institutions that could offer education to the general public as they helped the general public to "better themselves and appreciate the value of modern life" (p.4). Different from schools where the general public received formal education, museum architecture was understood to be "the advanced school of self-instruction" (p.5) and offered opportunities for the general public to conduct self-directed and selective learning. However, this idea was difficult to achieve because of the divergence of the overall educational role of schools and museums during that particular period in history. In addition, the new generation of curators was more interested in the accumulation of collections rather than in the public use of museums (p.5).

In the last three decades, the educational role of museum architecture has become venerable and notably because "the very nature of education in the sense of what we mean by the term and what we expect of educational institutions has changed" (Hein, 1998, p.6); learning is not to be achieved by means of written words in the traditional sense but should be "viewed as an active participation of the learner with the environment" (p.6). During the time that

the educational function of museums has developed, their other role as "interpreters of cultures" has been brought to the foreground in museums (p.9). Social theorists, such as Sharon Macdonald and Gordon Fyfe, argued that museums as social and cultural sites can create interest, such as "the stories museums tell, the technologies museums employ to tell stories, and the relation these stories have to those of other sites" (Macdonald and Fyfe, 1996, p. 3). What is learned in the museum and how learning takes place by interpreting cultures in the museum are more significant than a matter of intellectual curiosity (Hein, 1998, p.12). Therefore, the recognition of combined educative and interpretive roles supports the social function of museum architecture in the contemporary world.

In the modern age, in addition to their original function as "cabinets of curiosities" demonstrating personal collections, the accepted meaning of architecture of museums can be regarded as an "artificial memory, a cultural archive" (Hein, 1998, p.8), which has to be created in the pursuit of "historical memories recording by books, pictures, and other historical documents" (p.8) for modern humans to define and better themselves and to appreciate the value of modern life. Collecting and creating an archive of artifacts would ensure that tangible artifacts "would be saved from destruction through time by the technical means of conservation" (p.8) not only in the practical sense but also in the ideal sense; that is, the significance of tangible artifacts would be conveyed by interpretation and inherited by building up an engaging environment for self-learning.

The functional shift of museum architecture from storehouse to common place for communication also results in a functional shift of museum curators. Unlike the traditional function of the keeper of collections, to curate in the postmodern era means to "mobilize collections, to set them in motion within the walls of the home museum and across the globe as well as in the heads of the spectators" (Huyssen, 1994, p.21). With the social significance of constructing and facilitating a "self-learning" environment, museums adopt storytelling in their daily display routine to communicate the meanings contained by significant artifacts.

2.2. From oral to spatial

Museums have inherited the custom of telling stories to perform activities on their own. Deborah Mulhearn, a freelance journalist, reported that "oral history has come a long way in museums" (Mulhearn, 2008, p.29), which could be viewed as a treasured link with the past and a prominent way of "recording lives and unexpected events that may otherwise have been lost" (p.29). According to Leslie Bedford, a member of the Museum Group and the principal of Leslie Bedford Associates New York, many museums and historic sites rely on employing professional people to tell stories about a world that no longer exists. For Bedford, art museums particularly adopt storytelling to help all age groups communicate with collections; for example, the Art Institute of Chicago hosts parent workshops to teach them how to "read" art as a visual story and then takes them into the galleries to practice and refine the technique. According to Bedford, the oral usage of storytelling applied in the museum offers visitors opportunities to connect their experience of adventures or events to the museum environment, "which distinguishes programs from story hours at the local library for instance" (Bedford, 2001, p.32).

In addition to oral usage, storytelling has been applied in museums by use of exhibition techniques. One of the examples selected by Bedford is an exhibition strategy called "object theater," which was developed in the 1980s. By creating a multimedia and multisensory context with computer technologies, "object theaters" are designed in museums to "bring objects to life without necessitating a hands-on experience" (Bedford, 2001, p.29). According to Bedford, a successful object-theater experience is an exhibition at the Minnesota Historical Society, USA, as it consists of a physical set with a piano of various framed family photographs, a suitcase with a man's coat folded on top, and a kitchen table with a birthday cake, indicating the moments of loss and change in the human cycle of life. Framed by the song "Everything must change," universal experiences "resonate deeply and emotionally with visitors" (p.30), beginning with a home film of a toddler blowing out his birthday candles and ending with an elderly man doing the same, along with every person shown in the film telling their story as appropriate photographs or other artifacts are lit up. Bedford also indicated that historic museums adopt storytelling in their exhibition strategy by use of diverse forms of labeling to build up engaging exhibitions and thus "give voice to people and communities previously left out of the historical record" (p.31) to transform traditional exhibitions into live and vivid stories to be told and experienced.

From the above-mentioned discussion, storytelling is found to transform from its original verbal usage into diverse exhibition techniques and not only promotes communications in a verbal sense but also helps reconstruct a physical space to generate experiences. By participating in cognitive activities of storytelling, the general public is invited to "imagine another time and place, to find the universal in the particular, and to feel empathy for others" (Bedford, 2001, p.33). Storytelling is achieved at verbal and exhibition levels. The following question arises: how does storytelling as an embodied architectural space to connect individual artifacts from different times and places communicate with the general public individually and collectively?

For Tony Bennett, Professor of Cultural Studies and Foundation Director at the Institute for Cultural Policy Studies in the Faculty of Humanities, at Griffith University, Australia, museum architecture is "the specific cultural institution which requires a distinctive architecture of its own in the late eighteenth century" (Bennett, 1995, p.181). According to Bennett,

> The museum, as "backteller," was characterized by its capacity to bring together, within the same space, a number of different times and to arrange them in the form of a path whose direction might be traversed in the course of an afternoon. The museum visit thus functioned and was experienced as a form of organized walking through evolutionary time. (p.179)

With reference to similar or different messages contained by artifacts, early collections in public museums are usually classified geographically or chronologically. Storytelling is generally considered a context of viewing collections

visually, haptically, and kinesthetically and a linear spatial arrangement in the form of routes that the visitor is "expected – and often obliged – to complete" (Bennett, 1995, p.179). According to Andrea Witcomb, Associate Professor of Faculty of Arts and Education at Deakin University, Australia, the spatial arrangement of collections in many museums by use of linear narrative comprises three levels (Witcomb, 1994, p.240):

1. Level of space, in which the linear narrative is designed as "a one way flow, with exhibits lining either side of the rectangular space and a tunnel through which visitors must pass."
2. Level of collections, in which artifacts are usually set within another single linear narrative, "such as an evolutionary chronology from primitive to modern."
3. Level of individual artifact, which is "organized in a linear fashion, replicating the master narrative in the way it is classified, labeled, and displayed."

Witcomb argued that an authoritative strong linear narrative is reflected and produced by a combination of three levels of narratives, which can reduce various meanings of artifacts to only one point of view; that is, either that of the curator or that of the institution. As a result, visitors are thus "unambiguously placed as a receiver of knowledge, as the end-point of the production of the process" (p.240). For Witcomb, a link need not be established between various general or specific meanings contained by artifacts in many museum cases; that is, "displays could stand on their own and had no necessary connection on either side of them" (p.244), which can offer individual artifacts potential spatial positions in which to be arranged. Thus, the traditional linear narrative between displays, which is achieved thematically, chronologically, or by artifact type, is challenged. The spatial arrangement of authoritative narrative is also shifted from an overall spatial level to a detailed artifactual level and from a general to an individual level.

In summary, the evolution of storytelling in museums undergoes four stages: oral, exhibition, spatial, and artifactual. When storytelling as a spatial technique is applied in public museums, it transforms from an authoritative linear narrative of spatial arrangements of collections to the increasing attention on displays of individual artifacts. Therefore, museum professionals should rethink on the potential significance of artifacts by use of diverse storytelling design techniques to convey meaning and engage museum visitors. When applied in a museum environment as a mediator, storytelling should contribute to the promotion of multiple communications instead of authoritative "reading" and facilitate "an environment where visitors of all ages and backgrounds are encouraged to create their own meanings and find the place, the intersection between the familiar and the unknown, where genuine learning occurs" (Bedford, 2001, p.33). In Section 3, the significance of museum architecture as a medium of spatial storytelling will be interpreted with categories of spatial layout and circulation, spatial form and symbolization, and the spatial qualities of lighting and material, to see how a primary example of Jewish space is manifested spatially and cognitively in the YVHHM.

3. Spatial storytelling: The Yad Vashem Holocaust History Museum

3.1. Preface

Modern Jerusalem has developed outside the old city as it is located in the Judean Mountains, between the Mediterranean Sea and the northern tip of the Dead Sea. According to Leah Garrett, Professor of Contemporary Jewish Life and Culture at Monash University, Australia, "the Jewish people have always had a unique conception of both landscape and geography" (Garrett, 2003, p.108). For most Jewish people, life in exile before the establishment of the State of Israel in 1948 resulted in a deep recognition of the sacred notion of "home," which differs greatly from that of non-Jews in two ways: the real physical space where one lives in the present; and the spiritual imagined place of the mythical locale of the Land of Israel, which is often called "Eretz Yisrael" and refers to the biblical Land of Israel (pp.110–112). Garrett argued that the Holocaust was not merely a genocide, which wiped out Jewish people from where they lived physically, but radically altered Jewish descendants' conceptions of "home" (p.116), because "contemporary Jews as a whole have become less religious, more secular, and more interested in embracing the surrounding culture as well as being more accepted by many other nations" (p.118).

Yad Vashem is adjacent to the forest of modern Jerusalem and located on the hilltop of the western slope of Mount Herzl, also known as the "Mount of Remembrance." According to the online *Oxford Dictionary*, "Yad" (from Modern Hebrew yāḏ) means "monument or place" (in biblical Hebrew means "hand or arm"), "va" (in Hebrew wā) means "and," and "shem" (in Hebrew šēm) means "name." Avishai Margalit, Professor Emeritus in philosophy at the Hebrew University of Jerusalem, argued that the original name of "Yad Vashem" is based on Isaiah 56:5, "Even undo them will I give in mine house and within my walls *a place and a name* [Yad Vashem] better than of sons and daughters: I will give them an everlasting name that shall not be cut off," which indicates the promise of the Lord to offer a memorial in His city, Jerusalem, to the pious eunuchs (or castrated men), who would unable to have children but could still live for eternity as their names would survive long after them (Margalit, 2002, p. 22). According to Margalit, in September 1942, Mordechcai Shenhabi, a member of a secular kibbutz, suggested setting up a memorial under the name Yad Vashem for the Jewish people murdered in Europe, although most of people who were to become Holocaust victims were still alive at that time (p. 22).

Established in 1953 as the Holocaust Martyrs' and Heroes' Remembrance Authority, with the support of Jewish communities and organizations worldwide, Yad Vashem leads the historic mission to memorialize every Jewish victim of the Holocaust by collecting "Pages of Testimony," consisting of approximately 46,000 audio, video, and written testimonies by Holocaust survivors. By naming the memorial for the Holocaust victims, the sacred landscape of Yad Vashem expresses the idea that "the Jewish victims in Europe are like the eunuchs who leave no trace, and that there will be a national depository for their names, on the model motioned

in Isaiah" (Margalit, 2002, p.22). Opening to the public in March 2005, the dedication of the YVHHM was the culmination of a redevelopment project for 10 years, designed by world-renowned Canadian-Israeli architect Moshe Safdie.

The landscape of Yad Vashem plays a unique role in the formation of Jewish history and culture and makes the YVHHM a primary example of Jewish space with a specific engagement with *historical time*, with which Holocaust museums in other parts of the world cannot compete, such as the Ann Frank House in Amsterdam, the United States Holocaust Memorial Museum in Washington DC, or the Jewish Museum in Berlin. Thus, the YVHHM as spatial storytelling of *historical time* will be analyzed in this section as a visiting route with three narrative themes, namely, the Journey Home, the Dark Years, the Present and Future, with categories of spatial layout and circulation, spatial form and symbolization, and the spatial qualities of lighting and material throughout.

3.2. Journey Home

Approximately four miles past sprawling campuses of museum, government, and educational institutions at Givat Ram (a neighborhood in central modern Jerusalem), a signpost points to the right, and the Journey Home starts at the security checkpoint of Yad Vashem. Located in the forest valley below Yad Vashem, a large transportation drop-off area accommodates arriving buses, cars, and pedestrians at the mid-level of the mountain (Fig. 1a). Existing roads are integrated into a series of landscaped terraces; thus, the general public is required to climb the mountain from this point to the museum above and walk alongside a pine forest with asphalt and traffic on both sides. The spatial layout of this part consists of four architectural elements: the *piazza*, the *aqueduct*, the *mevoah*, and the *bridge* (from bottom right to top left of Fig. 1a). The *piazza* as an extension of the drop-off area for visitors that gather and look around fans out at 45° in deference to the topography (Fig. 1a). In addition, two visible upright architectural elements are present on the *piazza*, namely, a giant concrete wall and a box-like construction (Fig. 1a).

The *aqueduct* is a giant, smooth, and hollow concrete screen wall penetrating into the blue sky and consists of 12 big archways (Fig. 1b). With an inscription from the prophecy of Ezekiel on the aqueduct, "I will put my breath into you and you shall live again, and I will set you upon your soul," this concrete monolithic architectural element is not only an impressive door that links the *piazza* and modern Jerusalem but also an extraordinary topographical sign in

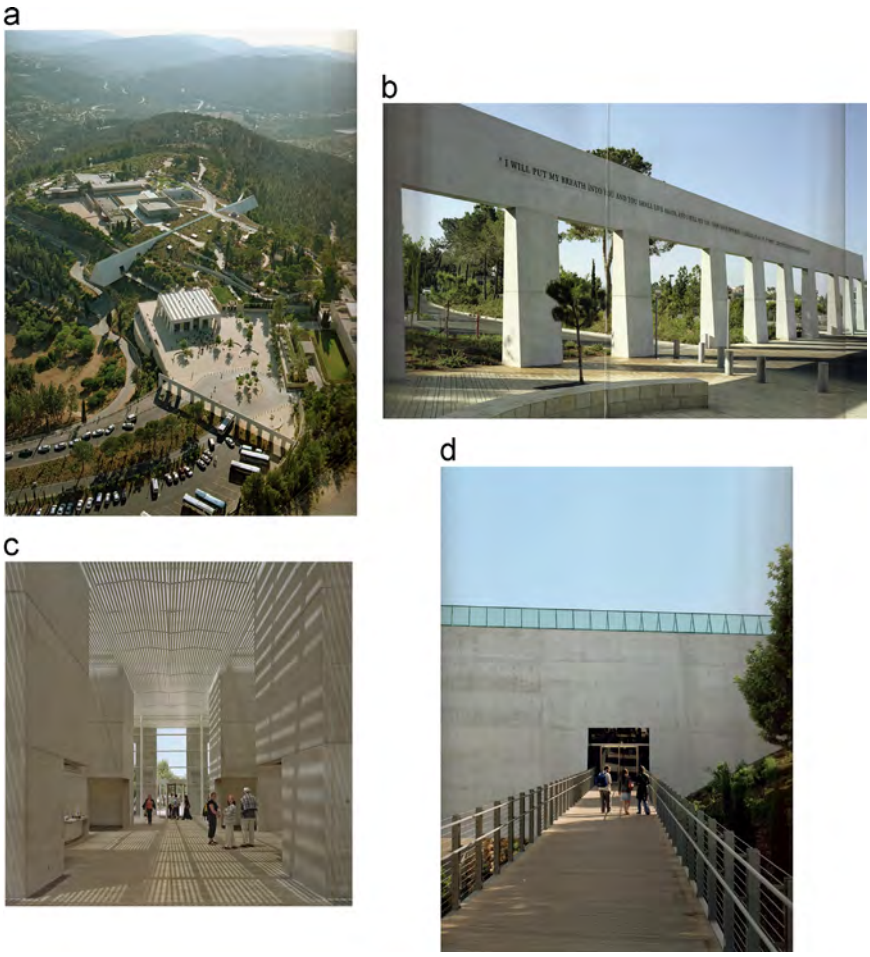


Fig. 1 (a) Northeast bird's-eye view of museum campus, from bottom right to top left, respectively: transportation drop-off area, aqueduct, piazza, mevoah, bridge (p. 3); (b) aqueduct (p. 8); (c) interior of mevoah (p. 9); (d) view of visitors entering bridge (p. 35).

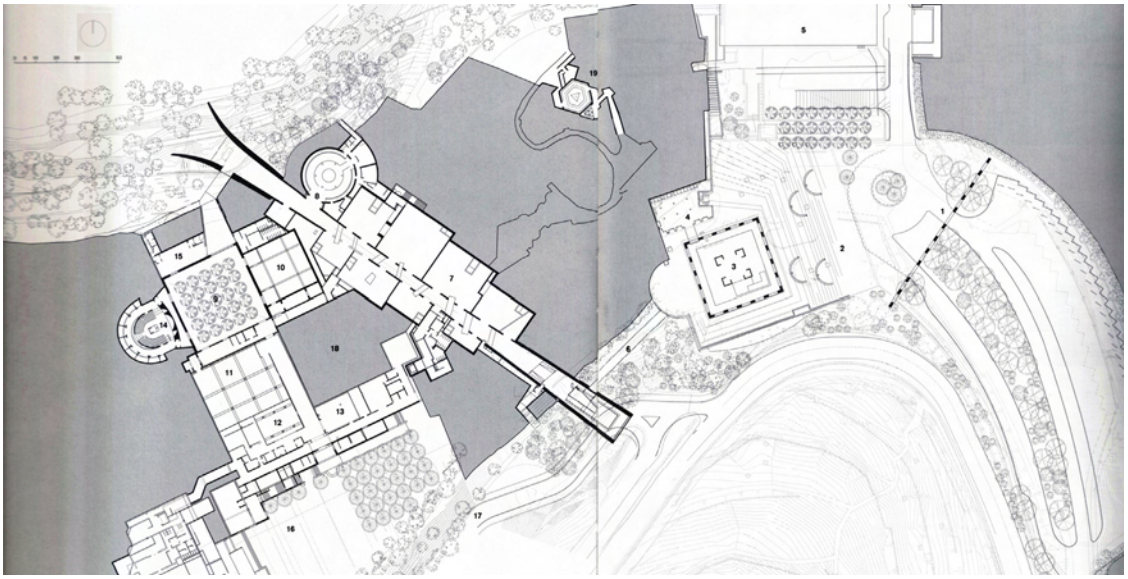


Fig. 2 Museum campus plan: 1 gateway wall; 2 entry piazza (underground parking below); 3 visitor centre (mevoah); 4 museum shop; 5 administration building; 6 entrance bridge; 7 Holocaust History Museum; 8 Hall of Names; 9 courtyard; 10. museum of Holocaust art; 11 exhibitions pavilion; 12 visual centre; 13 learning centre; 14 synagogue; 15 café; 16 Warsaw ghetto square; 17 avenue of the righteous among the nations; 18 Hall of Remembrance; 19 children's Holocaust memorial (p. 29).
Source: Safdie (2006a, 2006b).

scale and symbolic intent. At nearly eight times the height of an ordinary person, the scale of the columns further emphasizes the prominence of the sacred site where it opens toward the YVHHM. Edged on four sides with concrete screen walls and roofed by a delicate aluminum trellis, the *mevoah* is a box-like square pavilion that houses reception and information facilities in a visitor centre; this part is not included within but is rather excluded from the main museum space (Fig. 1a). The design of the trellis makes its interior space a lacework of dark and light lines, thereby allowing the changing shadows of the Jerusalem sun to form different patterns (Fig. 1c); the design also generates visitor speculations on the architect's intention to evoke the striped patterns on the clothing of the concentration-camp inmates (Safdie, 2006a, 2006b, p.96). As clarified by Safdie, "each person will create his or her own associations and symbolic interpretations, for architecture is not about prescribing what you ought to feel or think" (p.96). Rather than using local delicate and pretty sourced stone, concrete materials were selected in accordance with Safdie's concept of abstraction; that is, the feel of an archaeological remnant, whereby it slowly dawned on him that the aqueduct, the mevoah, the museum, and the auxiliary structures beyond must all be cast-in-place concrete (p.96). For Safdie, "only concrete could achieve a sense of the symbolic extension of the monolithic bedrock, free of joints, mortar, or any other embellishments" (p.96).

In addition to the *aqueduct*, the *piazza*, and the *mevoah*, the fourth architectural element designed on the Journey Home before entering the museum is a delicate steel *bridge* (Fig. 1d). Contrary to the solidity of other introductory architectural elements, the lightness and transparency of this bridge even result in it being ignored by visitors viewing it from a distance. The bridge with wooden-plank floors acts similar to the gangway of a liner waiting for visitors to board, which links the edge of the *piazza* to the museum: one is a functional, quotidian, real world; the other is a world that has once physically existed but has now vanished.

From the analysis of the above-mentioned elements, the circulation of the Journey Home is found to experience a rhythmic exploration, which inhibits visitors' impulse to enter into the museum straightforwardly. Replaced by a series of completely different but not disconnected anamnestic segments, that is, the *piazza* as physical extension of quotidian world; a piece of Jewish literary inscription in the *aqueduct*; illusions of concentration-camp clothes presented metaphorically on the *mevoah*, the architect's design techniques successfully foreshadow clues for the following route. From here, the level of the earth descends by degrees and the Dark Years await.

3.3. Dark Years

In response to topographical features and with respect to the sacred position of the Mount of Remembrance and the Yad Vashem campus, the museum space is presented entirely underground. Under normal circumstances, underground space rarely possesses satisfactory rather than auxiliary functions for its specific location without light and air, thereby easily producing feelings of cold and claustrophobia and requiring complicated construction and high costs. In this case, the design of underground space not only contributes to creating a harmonious atmosphere between the natural and built environments but also makes interior space of the museum mysterious and unpredictable.

As shown in the museum plan (Fig. 2), the spatial layout of the Dark Years consists mainly of a narrow, central, and linear corridor with asymmetric galleries integrated on each

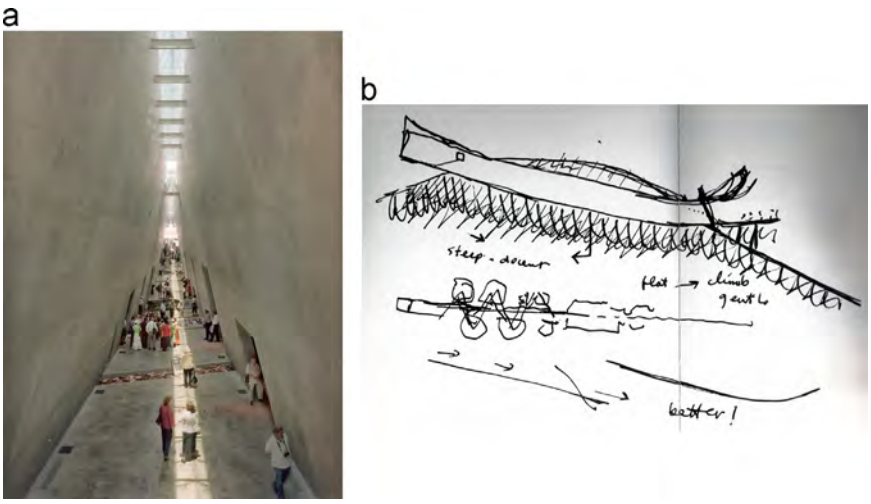


Fig. 3 (a) Interior view of lighted-filled central corridor, with visitors walking and talking. Barriers, filled with artefacts can be seen on the floor (p. 45); (b) Safdie's sketch indicates his original conceptions of museum space in relation to sacred site (pp. 86f). *Source:* Safdie (2006a, 2006b).

side; that is, four to the north-east and five to the south-west. Upon walking into the museum, the central structure is found to be designed as an extreme, gigantic, triangular, prism-like, and linear walkway and is around 300 m long, 30 m high, and from 6 m to 12 m wide (Fig. 3a). Sketched conceptually by Safdie and realized in reality (Fig. 3b), this three-dimensional form cuts through the Mount of Remembrance from the south, extends underground, and emerges and explodes to the north, with both ends cantilevering dramatically into the open air (Fig. 1a). Morphing in the viewer's imagination from a primitive shelter or tent, to a half Star of David, a fir tree, and an ark filled with survivors, to the ominous pitched roof of a train shed full of deportees, or a gas chamber at Treblinka (Ockman, 2006, p.23), the decision for a wedge-like, prismatic, and organic shape depends on its primitive and native character to maintain the gravitational force rather than any symbolic purpose (Safdie, 2006a, 2006b, p.96). Although underground, the central walkway is drenched with sufficient natural light, which enters through an overhead, triangular-shaped, and glass skylight (Fig. 3a).

Unlike the Journey Home as a free and leisurely visit, the Dark Years can be viewed as a visit that visitors are expected and obliged to complete; this journey starts at an inescapable 10 min video art display, that is, *The World That Was*, which portrays the Jewish world before the Holocaust, installed in a congested space around the south gable wall (Fig. 4a). With images and sounds, this montage invites visitors to sense the former days that these people once experienced with joy and sadness, difficulties and deliberations, and exploration and education. This place symbolically becomes the starting point of the Dark Years and the opening page of this story. As visitors turn around, they will be drawn by large photograph exhibits on both sides of the prismatic walls, with eye-level installations on another upright layer of glass screen in front of them (Fig. 4b). This display area, broken by an embedded and enclosed glass terrace of the outdoor area, shows the museum design group's ingenious arrangements for the second stop in circulation. Contrary to the shocking piles of corpses of Jewish prisoners killed in a camp in Klooga, Estonia, as the Red Army advanced in September 1944, depicted in the wall-sized photographs on a solid concrete background, normal-sized charred photographs found in the victims' pockets, reflecting their life stories and printed on an upright glass screen in the foreground, reveal a sense of transparency and physical fragility. According to Avner Shalev, Chairman of Yad Vashem, the intention of presenting the mass murders at Klooga at the opening of the visit was not because it was an exceptional event in Holocaust history but was rather an introduction to the layered structure of the exhibit and to the narrative related in the museum (Shalev, 2006, p.95). Consisting of mixtures of different textures, namely, transparent glass, paper plate, and concrete integrated with natural and artificial lightings, this spatial introduction effectively encourages visitors' perception of distant *historical time* and their gradual involvement in this particular conversation.

As visitors proceed along the central walkway straddling the prism, galleries reveal themselves slowly as a series of interconnected and compacted square chambers of various size that wait for visitors to discover them (Fig. 2). Even with large openings at their entrance, the galleries are significantly less bright and even dim as they rely on warm-toned artificial light and minimal skylights contrary to the central space that is richly illuminated (Fig. 4c). Safdie mentioned that the inspiration for chamber galleries was the ruins of Qumran, along the Dead Sea's north-west shore in Jordan, where the Dead Sea Scrolls were found (Oppenheimer, 2005, p.114). Compared with the cultural significance contained in the scrolls as significant archaeological discoveries, Safdie articulated these galleries as giant burial caves in the Mount of Remembrance (Safdie, 2006a, 2006b, p.95). Analogous to chapters of a book, each gallery has a title: From Equals to Outcasts – Nazi Germany and the Jews; The Awful Beginning – World War II and the Beginning of the Destruction of Jewish Life in Poland; Between Walls and Fences – The Ghettos; Mass Murder – The "Final Solution" begins; The "Final Solution" – Jewish Uprisings in the Midst of Destruction; Resistance and Rescue;



Fig. 4 (a) Video and art display of 'The World that was', with visitors standing in front watching and thinking (p. 38); (b) interior corridor space with multi-layered photo displays, above which an exterior terrace can be clearly seen (p. 37); (c) interior view of lighted-filled central corridor, with barriers filled with artefacts and large entrances to galleries on either side (p. 40); (d) interior space of Gallery 5 – The 'Final Solution': Jewish Uprisings in the Midst of Destruction – visitors are looking at the floor display of victims' shoes (p. 67); (e) interior space of Gallery 7 – The Last Jews: The concentration camp universe and the death marches – visitors are looking at window display of victims' camp clothes (p. 47).
Source: Safdie (2006a, 2006b).

The Last Jews – The Concentration Camp Universe and the Death Marches; Return to Life – She'erit Hapleita. With each gallery dedicated to depicting the Holocaust by chapters in a chronological and thematic course, the entire story is told from a Jewish perspective through the linear circulation of the galleries, combined with personal stories of 90

Holocaust victims and survivors, which are presented in exhibitions of 2500 personal items, including artwork and letters from the Holocaust donated by survivors and others.

To discourage visitors from taking shortcuts between galleries, visitors' circulation is intentionally arranged as a zigzag way into adjacent galleries by way of a series of impassable barriers in the museum plan (Fig. 2). Unlike the meticulously light-filled positions in the linear space floor, monstrous artifacts are displayed in these barriers, which function metaphorically as the thematic titles of each gallery. Installed with victims' artifacts, such as books and even burned railway tracks together with textual exhibition techniques (Fig. 4c), these barriers serve as chapter headings for an evolving historical narrative of the exhibitions within galleries and as symbolic historical turning points to remind visitors of important events in the Holocaust. Arranged in a rhythmic circulation between the galleries and the central corridor, these barriers as transitional and buffering areas between galleries also offer visitors opportunities to review the stories learned in the previous gallery and to imagine what stories will be learned in the next gallery. By tying historical themes symbolically to a tight and concise spatial sequence, the central corridor thus plays an irrefutable role in mediating visitors' understanding of what really happened to the Jewish people and the world during the Holocaust, thereby revealing, interpreting, and mediating *historical time*. As argued by Merleau-Ponty, "our perception is inextricably bound to movement" (Locke, 2016, p.4). As visitors walk along, their perception of *historical time* is bound inextricably to their bodily movement, along with their awareness of horizons. Rather than simply a backward-looking process of identifying *historical time* "behind" the artifacts, this linear space as "a finite, subjective-relative world with indeterminately open horizons" (Gadamer, 1992, p.193) reshapes *historical time* "in front of" the artifacts, that is, the ways of life made possible by it.

In addition to the deliberate asymmetry along the corridor, scales and displayed manners between galleries vary owing to the content of displayed artifacts (Fig. 4d-e). Compared with the symbolic, dominant, and embodied experiences created by architectural space, the artifacts in each gallery contribute to illustrating individual stories with entire themes and events and bring out the human dimension more than ever before. Moreover, by means of creating eye-level encounters between the "narrator" or "witness" within each gallery, visitors are led through a two-tiered exhibit structure: the context, which describes the historical processes, and the artifact, which tells the story at a personal level (Shalev, 2006, p.54). In concert with spatial techniques, the complexity of the events necessitates a chronological and thematic approach; the entire visiting route is designed from spatial to artifactual, from outline to detail, and from collective to individual.

Upon lingering in the corridor, the warped surface of prismatic structure is found formed and amplified by a gently sloping floor; in particular, the floor and roof planes ramp five degrees downward and then stop, which creates a changing sequence of spaces and gives the expectation of descending deep into the mountain, with the exit bursting forth from the slope of the mountain to a dramatic view of modern-day Jerusalem (Fig. 3a). As explored by visitors as they move from one gallery to the next and from one chapter to another, the central space is therefore to be understood "not only through the eyes and mind but also through the more sensory experiences, with strong visceral feelings" (Tuan, 1975, p. 152). Owing to the rhythmic play of strong natural light and dim artificial light between central space and galleries, an extraordinary experiential dimension within space and place is created and constructed. The interweaving lighting scheme offers visitors an overall impression of time, place, and atmosphere, between darkness and light, between fiction and reality, between temporality and permanence, and between the past and present. The central linear corridor shelters space and life and not only successfully creates a visiting experience full of visual and spatial variations but also acts as "a shelter, a sarcophagus; in short, a form of embodiment" (Ockman, 2006, p.24). The museum is not only "ideally equipped to deal with the surplus of history that it would order, store, and display" (Giebelhausen, 2003, p.1) but is also a legitimate part of a great culture that can be drawn by architects. By means of the integral storytelling of artifacts and architecture, visitors' understanding of the Holocaust is elaborated not only at the artifact and exhibition levels but also in terms of embodied and architectural space.

3.4. Present and future

The Hall of Names is located at the north end of the east side of the corridor, in a break from the orthogonal, linear, and square geometries of the museum plan (Fig. 2) and is circular and surrounded by polygonal areas. The Hall of Names serves as an archive and a memorial and is a repository of the Pages of Testimony submitted by survivors, relatives, and friends; in other words, the part is a memorial to the six million Jews who perished and an especially revered place within the sacred site of Yad Vashem (Safdie, 2006a, 2006b, p.98). Unlike any of the previous galleries, a buffering space is designed between the central walkway and its interior space for visitors to stand and read the introduction, thereby making it a significant threshold with depth (Fig. 5a).

Upon entering through an elevated and ring-shaped platform (Fig. 5b), the main area is found to be composed of two reciprocal cones, which are constructed nearly the same as Safdie's original concepts (Fig. 5c-d). With one towering 10 m into the sky and a reciprocal well-like one excavated 10 m into the bedrock of the Mount of Remembrance, the base of which is filled with water; visitors are offered the best position to see the interior conical spaces by standing on the platform surrounded by transparent glass balustrades (Fig. 6c). The upper cone, designed by exhibition designer Dorit Harel, provides a visual display and features 600 photographs of Holocaust victims and fragments of the Pages of Testimony (Fig. 6a). The lower cone acts as a quiet mirror reflecting and memorizing people whose photos appear above and whose names will never be known (Fig. 6b). The polygonal areas surrounding the Hall of the Names form the repository, house approximately 2.1 million Pages of Testimony collected to date, and have empty spaces for those yet to be submitted (Fig. 5b). With this

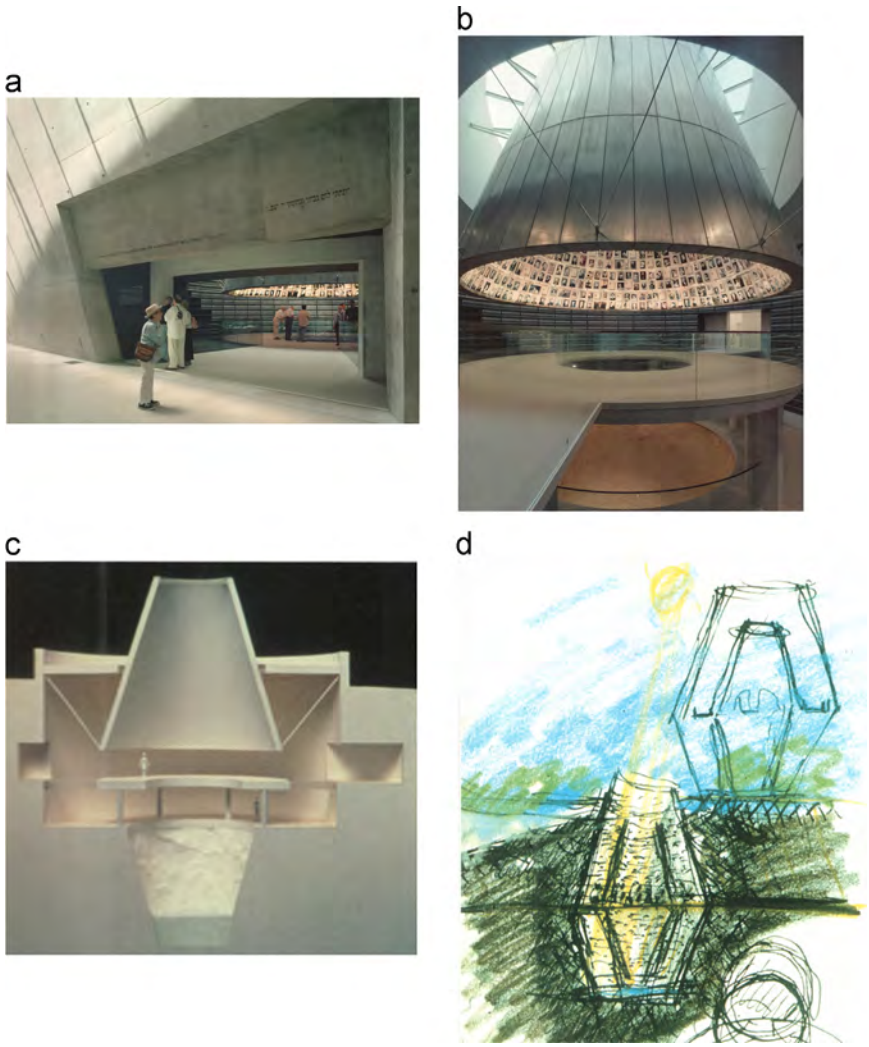


Fig. 5 (a) View of entrance to Hall of Names (p. 80); (b) interior of Hall of Names, viewed from the platform. Surrounding areas are collections of Pages of Testimony collected so far, with empty spaces for those yet to be submitted (p. 81); (c) referential physical sectional model of spatial configuration of Hall of Names (p. 80); (d) Safdie's sketch shows conceptual design of this room (p. 90). Source: Taken from Safdie (2006a, 2006b).

salvaged installation, visitors are reminded of how many names are still missing (Shalev, 2006, p.60).

In reference to the exceptional spatial design, particularly regarding the form of this room, natural light is invited in through a glass-covered skylight above the repository (Fig. 5b). The upper cone relies on warm-toned artificial light and faint natural light, heralds the climax of the display, and manifests the spatial prominence of it. The victims peer at visitors as if they are looking for a response, thereby inspiring the visitors to reflect on a personal responsibility (Fig. 6a). This brilliant presentation guarantees that "the distant past resounds with echoes, and it is hard to know at what depth these echoes will reverberate and die away" (Bachlard, 1994, p.xii). According to one curator, Joan Ockman, some drawbacks to the final space can be observed:

> It is unfortunate, that the photos and Pages of Testimony reproduced in the heaven of the dome are too high to read. Compared with the photos of faces within the cone, the binders are faceless. They appear more a decorative device than an archival resource. Apart from this, the collaboration and counterpoint between curatorship and architectural design remain admirably balanced and harmonious. (Ockman, 2006, p.25)

Even with limitations, visiting this space potentially generates two modes of experience for visitors: one is resonance, that is, "the power of the displayed object to reach out beyond its formal boundaries to a larger world, to evoke in the viewer the complex, dynamic forces such from which it has emerged and for which it may be taken by a viewer to stand" (Greenblatt, 1991, p.42); the other is wonder, that is, "the power of the displayed object to stop the viewer in his or her tracks, to convey an arresting sense of uniqueness, to evoke an exalted attention" (p.42) (Fig. 6c). Regardless of which mode of experience is encountered by visitors, exhibition and architecture are welded together extraordinarily in the most successful space within the museum; such arrangement provides visitors a dramatic visual collision between the individual and collective, between space and place, and between

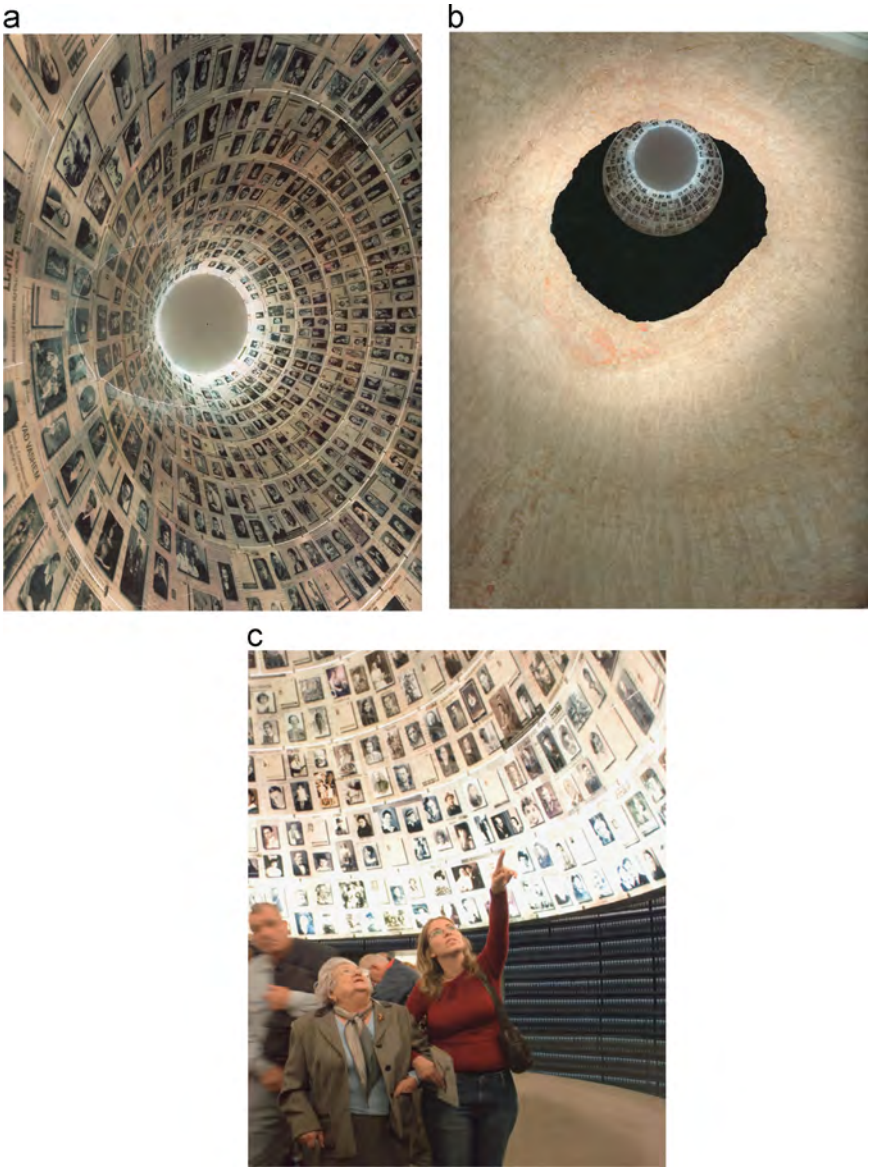


Fig. 6 (a) Interior of upper cone of Hall of Names with shocking photo montage (p. 83); (b) interior of lower cone of Hall of Names (p. 85); (c) view of interior of Hall of Names, visitors are trying to speculate on recognition of faces shown above (p. 82). *Source:* Safdie (2006a, 2006b).

artificial and natural. In addition, these components evoke visitors' empathy for an unknown Jewish person and for all those murdered in the Holocaust, thereby establishing "deeper connections between museum artifacts and visitors' lives and memories" (Bedford, 2001, p.30). Memory is the shell of that was once a body, a form of motionless disembodied reflection (Ockman, 2006, p.24), and is tangibly shaped in the embodied experience of this architectural space; the more securely the memory is fixed, the sounder it is (Bachlard, 1994, p.9).

The story of the Jewish people comes to an end at a stunning terrace (Fig. 7a), which consists of the north wings of the central prismatic walls and images of the past behind, which unfold into the magnificent landscape of the Jerusalem hills ahead (Fig. 7b). This architectural *mise-en-scène* with the landscape emphasizes "the quality of light, the question of social visibility, the power of the eye to naturalize the rhetoric of national affiliation and its forms of collective expression" (Bhabha, 1990, p.295). With a vivid sense of living and existing, it brings visitors from the dark into the light and from the past into the Present and Future.

> Years before, designing the Children's Memorial had given me an inkling of the power of emerging into light. It meant that life prevailed. For the new museum, cutting through the mountains and bursting northward, dramatically cantilevering the structure over the Jerusalem pine forest provide views of the hills beyond took this life-affirming experience to another level. To stand on the extended terrace, the side walls of the prism curving away from the site seemingly into infinity, and see the fresh green of the recently planted forest with its great sense of renewal and the urbanizing hills beyond is to



Fig. 7 (a) Interior view of terrace, looking outward (p. 105); (b) exterior view of terrace in forest below (p. 109). *Source:* Safdie (2006a, 2006b).

> understand that, indeed, life prevailed. We prevailed (Safdie, 2006a, 2006b, p.99).

The circulation of the museum campus ends at a sunken paved courtyard, which is enclosed by solid walls and has local limestone on the ground that is planted with grids of trees and is open to the sky. According to Safdie, this courtyard was conceived as a decompression chamber and is the inverse of the *mevoah* through which the journey begins (Safdie, 2006a, 2006b, p.100) (Fig. 2).

Crafted throughout entirely in concrete, each part of the museum campus becomes a powerful corporeal element that exerts topographical influence on the sacred site. As the only tangible medium that can be seen, sensed, and touched by the general public, the concrete material validates architecture as a spatial embodiment of human beings' lives in the physical sense, without which other mediums of spatial storytelling will find no place to take root.

> From the Mount of Remembrance in Jerusalem, the significance of Yad Vashem thus continues to imbue the memory of the Holocaust with depth and meaning and to ensure that the memory of the victims and the voices of the survivors will resonate for all generations, from here, the overall story of the Holocaust is disseminated to the whole world – not only in its meaning for Jewish continuity but also in its universal significance (Safdie, 2006a, 2006b, p.63).

4. Conclusion: Significance of spatial storytelling

By telling the events of the Holocaust with a beginning, a middle, and an end and are laid out along a path that visitors can follow, the museum building successfully establishes an architectural dimension that displays, reveals, and interprets the untouchable *historical time* of Jewish people during the Holocaust. By means of spatial layout and circulation, spatial form and symbolization, and the spatial qualities of lighting and material, the sequential nature of such spatial configuration allows museum curators "to design exhibitions in accordance with the Holocaust's still-developing historical narrative" (Forty et al., 2006, p.34). Described by Safdie as "a volcanic eruption of light and life" (Forty et al., 2006, p.34), the museum validates itself further as a prominent metaphor of landscape by the achievement of its material consistency and lighting and material as mediums, which transform in the course of time. By generating dialogues throughout the prism form, by piercing the innards of the mountain without losing contact with the outside world, by arranging chamber galleries branching off both sides of the prism in a zigzag pattern, leading to unexpected displayed content, and by the reciprocal conical space of the Hall of Names, the museum is imbued with a sense of power that reaches its climax at the end of the visit, that is, architecture offers artifacts a physical context within which to "speak" for themselves. Artifacts, which encompass small artifacts contained within, and large architecture as fabric shelter construct a new layer of place.

Architecture as spatial storytelling not only contributes to reconstructing *historical time* by use of its space but also potentially creates various modes of experience for museum visitors. Kinesthetic experiences throughout the entire movement of circulation. Visual and haptical encounters are created by contact with the spatial qualities of uniformity of concrete materials and exquisite lighting design. Conceptual and mental experiences are symbolized and revealed by concise but potent geometric forms, that is, "to experience is to learn; it means acting on the given and creating out of the given. The given cannot be known in itself. What can be known is a reality that is a construct of experience, a creation of feeling and thought" (Tuan, 2001, p.9). Place as a centre of meaning is therefore constructed

by experience (Tuan, 1975, p.153). In other words, if Yad Vashem was the first social element that brought significance to the religious, historic, and cultural meanings of the Mount of Remembrance and made it a sacred site, then this new museum elaborates and emphasizes those meanings architecturally, topographically, and symbolically. The experience of visiting a series of landscaped terraces and to "journey home," physically as well as metaphorically, constructs meanings of the Jewish "road": not only the "territorial" road, the physical path that connects where Jewish people live now and where the dead dwell in Yad Vashem, but also a symbolic, mythic road "as Galut intersected with the territorial road, making the Jew move through both historical and mythic-poetic time and place" (Garrett, 2003, p.114). The experience of visiting the Dark Years, particularly through the circulation of bounded galleries, reveals a communal Jewish narrative. The experience of visiting the Present and Future in the Hall of Names creates meanings of a Jewish "home" for the Jewish victims of the Holocaust, thereby finding their roots in a sacred place. According to Garrett,

> There is no other nation in the world that gives instant citizenship to members of a specific religion. For Jewish people, however, who regarded themselves as sharing a culture, history, and peoplehood beyond a religion, the "law of return" supports the idea that all Jews are a common nation. (Garrett, 2003, p. 118)

The landscape consisting of architectural features and nature represents a way in which Jewish people have signified themselves and their world through their imagined relationship with nature and in the ways they have underlined and communicated their own social role and that of others with respect to external nature (Cosgrove, 1984, p.15). The recurrent metaphor of landscape as the *inscape* of national identity emphasizes the quality of light, the question of social visibility, and the power of the eye to naturalize the rhetoric of national affiliation and its forms of collective expression (Bhabha, 1990, 295).

"We do not have any mental record of who we are until narrative is present as a kind of armature, giving shape to that record" (Abbott, 2008, p.3). Through storytelling by use of architecture and landscape, physical time becomes *human time* to the extent that "it is organized after the manner of storytelling" (Ricoeur, 1990, p.3). Storytelling is thus meaningful to the extent that it portrays the features of the temporal experience (p.3), which claims human beings' existence in one particular place at a particular time. Nevertheless, "human time," similar to the growth, history, and culture of every individual, collective, and nation, embodies human beings in the real and quotidian world. Architecture therefore becomes a part of landscape, a longitudinal section of culture, and a cross section of history. The significance of the YVHHM not only lies in its creation as one of the world's first Holocaust museums but also lies in its unique capability to interpret and mediate a primary example of Jewish space. By employing the spatial storytelling specificity of *historical time* in architectural design, artifacts, architecture, and landscape construct a tangible sacred and cultural artifact; such artifact inherits, preserves, and creates the place of Yad Vashem, the modern City of Jerusalem and the Nation of Israel and a holy and an ideal physical and spiritual "home" for Jewish people worldwide. Without wandering anymore, the souls of buried lives will rest safely and in peace, and the unknown images in the Hall of Names will finally return "home" to Yad Vashem in the near future. The design of the YVHHM recalls the historical narratives of the Holocaust while its narrative, which has always existed, melts into a time and place.

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